

Project

Algorithm

Phase 1: Implemented after phase 1.1 of implementation.

Phase 2: Implemented after phase 3 of implementation

Implementation

Phase 1: Two decks and those cards only (Fignor, Igor)

1.1 Archon (Play)

1.2 Reversal (Swap and play)

1.3 Adaptive (Play, Swap and Play, Bid and Play)

For the third round, the deck that won both rounds is bid on. The owner of the deck goes first, with a bid of 0 chains. Alternative bids until someone stops or until 24 chains. Everytime someone with n chains would draw cards during step 5 of the turn order they draw $\text{math.ceil}(n/6)$ fewer cards and reduce their chains by 1.

Phase 2: Three houses, any deck from those so all cards (Logos, Shadow, Dis)

2.1 Alliance

Each house from each deck can be used separately to form a new deck

Phase 3: All first expansion, all 7 houses, any deck.

Phase 1.1:

Keywords:

Action:

If the card belongs to the selected house, may exhaust card to activate the effect. May only use one Action/Omni on a card per exhaustion.

After Fight:

Triggers after a Fight is initiated and the Fight phase passes or is skipped.

Assault (X):

Triggers if the Attacker has this keyword in the Before Fight phase. Deal X damage to the Target by increasing their damage counter by X. If they are destroyed (their damage counter is greater or equal to their power), the fight does not proceed. If multiple Assault (X) exist, sum them.

Archive:

Puts a card from the specified location, or if not specified the hand, into the archive. Some abilities trigger when a card is archived.

Before Fight:

Triggers after a fight has been initiated but before the fight. Any effects triggered in this phase occur in the order chosen by the active player.

Controller:

The person who has control over the card. By default, this is the owner, but can be altered via Effects.

Destroyed:

When targeted by a Destroy effect or when the damage on a card is greater than its power, tag the card as Destroyed. Then resolve the specified Effect if any. Finally, place the card into its owner discard pile unless an Instead keyword takes effect. If other Destroyed effects take place during this process, resolve all Effects before discarding all cards. If a card is already tagged as Destroyed, it cannot be tagged again and any attempt to destroy or sacrifice it do not work, and its Destroyed effect does not resolve again.

Effect:

Trigger:

Either a keyword or if no keyword is specified, always in effect

Steps:

An ordered list of steps to take to resolve the effect

Exhausted:

Triggered when Action, Omni, Fight, or Reap is triggered, or when an Artifact or Creature is initially played. The card cannot be used for the remainder of the turn. At the start of a turn, all exhausted cards are readied, meaning their exhaustion is removed.

Hazardous (X):

Triggers if the Target has this keyword in the Before Fight phase. Deal X damage to the Attacker by increasing their damage counter by X. If they die, the fight does not proceed. If multiple Hazardous (X) exist, sum them.

House:

One of the factions of cards. For this project, we are only looking at the original 7, Brobnar, Dis, Logos, Mars, Sanctum, Shadow, and Untamed. Each has its own theme and special abilities. A player may choose one house at the beginning of their turn, and then they can play any card in their hand of that house and use creatures and artifacts in the play area of that house.

Instead:

Replaces a trigger when condition is met. For example, instead of destroying a card, you would perform the alternative effect.

Fight:

Exhaust a controlled creature as the Attacker. Choose an eligible creature you don't control as the Target. Trigger all Before Fight phase effects. If either creature is now dead, skip to the After Fight phase. Each creature deals damage to the other equal to their power (by increasing their damage counter by whatever their power is) and trigger all Fighting phase Effects. The active player chooses the order these effects take place. If the Attacker was not Destroyed, trigger all After Fight phase Effects.

Flank:

The first or last card position in the set of cards in the PlayArea. When playing a new card, this means the new card ends up in the first or last position.

Omni:

Regardless of selected house, may exhaust card to activate the effect. May only use one Action/Omni on a card per exhaustion.

Owner:

The person who had the card in their deck at the start of the game

Passive:

The effect with this trigger is always active.

Play:

Triggered when a card is played.

Put:

Move a card from one location to another. This does not trigger other effects such as Archive or destroyed.

Purge:

Remove a card from the game entirely

Sacrifice:

Destroy a card you control.

Reap:

Exhaust a creature. Gain 1 Æmber to your Æmber pool. Then trigger After Reap effects.

After Reap:

Triggers after a creature reaps.

Classes to Implement

EffectObject: superclass for types of effects:

DurationEffect: for lingering effects

RemainingDuration (or source of effect if infinite)

PlayerAffected

Conditional (can be null)

Name of variable affected

Value of variable affected

InsteadEffect: for doing x instead of y

RemainingDuration

PlayerAffected

Do

InsteadOf

TriggerEffect: do x when y

RemainingDuration

PlayerAffected

Do

When

Card:

Æmber on Play: when played, gives this much Æmber to the pool

Æmber Captured:

Default of 0. When the card leaves play, this amount is added to the enemies Æmber pool.

ID: id of the card

Image: image of the card

House: identifies which house the card belongs to

Name: the name of the card

Tags: Identifier(s) that may be targeted by Effects

Type: identifies what type of card it is

TypeObject: Superclass for specific types.

Action:

Has some number of Effects

Discarded after Effects Resolve

Artifact:

Placed on the field

Enters play exhausted

Has a controller

Has some number of Effects

Creature:

Placed on the field.

Enters play exhausted

Has a controller (a player)

Has some amount of power above 0

Has some Armor, 0 or above

Has some damage, starting with 0

May have some number of Effects.

Can Reap

Can Fight

May have some number of attached Upgrades

Upgrade:

Attaches to any creature

Modifies some of the Creature's info

Modified Effects are considered to be the Creatures'

Discarded with the Creature

Variables to implement

ActiveEffectList:

A list of all effects that last longer than simply resolving now and being done. Includes all elements in TriggerEffectList and InsteadEffectList, but not the CurrentQueue. Often, a step of the Effect will resolve and then be added to the list rather than being added directly.

Archive:

An unordered collection of cards. All cards in the archives are added to a player's hand after they select a house. Cards can only be added via effects. The owner of the archive may view the cards at any time.

CurrentQueue:

Contains the next steps and all effects that need to be resolved before doing the next step in the turn order, organized by priority according to when they need to be done. Implemented via priority queue. When multiple things tie, the current player picks the next item to resolve. Ties should generally only occur when multiple things are triggered by the same keyword. Otherwise, effects should resolve in the order they are triggered, Before keywords happen before After keywords, and so on.

Deck:

A set of three Pods, each with a different House. The cards in a deck are ordered, with the first card being considered the Top card and the last card being considered the Bottom card. Implemented with a linked list. The player fills the hand from this pile, starting from the Top. Unless authorized by an effect, cannot be viewed by either player. However, each player may view a list of all cards existing in either deck in an unordered state at the beginning of the game at any time. When a deck is empty and the player needs to draw, the discard pile is put into the deck and then shuffled.

DiscardPile:

An unordered collection of cards. A player cannot interact with this set unless via an Effect. May be viewed by either player at any time. Implemented with a stack.

InsteadEffectList:

A list of all effects that change x into y. For anything that can be affected, this list is checked before doing x to ensure that it is not supposed to be y instead.

GetFunctions:

An assortment of functions that get any specific number, such as the card draw modifier (after chains and effects), the number of cards that can be played that turn (usually infinite, restricted by effects), and so on. Anything that can be affected by effects that are not immediately resolved or differences in the play state should be implemented as get functions so that everything is calculated when it is needed and effects resolve properly.

Hand:

An unordered collection of cards. If any card belongs to the currently chosen House, they may be used in any order. Can be viewed by the owner of the hand at any time.

PlayArea:

A pair of ordered collections, one of Creatures and one of Artifact cards, potentially with Upgrades attached to some of the Creatures. If any

Creature or Artifact belongs to the currently chosen House, or have the keyword Omni, they may be used in any order. Can be viewed by all players at any time. A player playing a new card to the PlayArea must place it on the Flank, but may choose whether the new card is first or last (if those are not the same).

Player:

One of two agents. Each has a play area, hand, discard pile, archive, purged zone, and deck. Also contains the variables listed in variables for the player.

Pod:

A set of twelve cards of a single house.

PurgedZone:

An unordered collection of cards. May be viewed by either player at any time. Can only be interacted with via Effects.

TriggerEffectList:

A list of all effects that trigger x when y is done. Checked before any y that could be a trigger to before adding to the CurrentQueue

Rule of 6: A given named card may only be played or used up to 6 times a turn. Copies share the same limit.

Turn Order:

1. Forge a key, if possible
2. Choose a House and set CanBeUsed of all cards you control of that house to True.
3. While there are cards that can be used/discarded:
 - a. List all available actions the player can take (any card that CanBeUsed and isn't overwritten can be used, plus any card of the selected house can be discarded). One additional action is to end the turn, forgoing any remaining actions.
 - b. Player selects one of the available options
 - c. Execute effects
 - d. Repeat
4. Change Exhausted to False for all cards you control, and change CanBeUsed of all cards you control to False.
5. Draw cards to meet the DrawUpToLimit.

After that the next player begins with step 1.

Setup:

Select a deck for each player.

Instantiate all objects for each player, including a shuffled decks, and Æmber, keys, and chains set to 0.

Deal 6 cards as the starting hand.

Player 1 may choose to mulligan their hand if they choose to (putting their hand back into the deck, shuffling, and then drawing 5 cards).

Player 2 may then choose to mulligan their hand if they choose to.

Decks

Fignor cards:

Dis:

- Arise
- Control the Weak
- Creeping Oblivion
- Gateway to Dis
- Three Fates
- Library of the Damned
- Dust Imp
- Shooler
- Snudge x2
- Succubus
- The Terror

Logos:

- Help From Future Self
- Labwork
- Phase Shift
- Scrambler Storm
- Sloppy Labwork
- Wild Wormhole
- The Howling Pit
- Doc Bookton
- Mother x2
- Timetraveler
- Titan Mechanic

Shadows:

- Bait and Switch
- Booby Trap
- Ghostly Hand
- Lights Out
- Miasma x2
- Nerve Blast
- Too Much to Protect
- Subtle Maul
- Noddy the Thief
- Silvertooth
- Urchin

Igor Cards

Dis:

- Arise
- Control the Weak x2
- Gateway to Dis
- Dominator Bauble
- Lash of Broken Dreams
- Lifeward
- Dust Imp
- Ember Imp
- Guardian Demon
- Pit Demon
- The Terror

Logos:

- Help From Future Self
- Labwork
- Library Access
- Sloppy Labwork
- Wild Wormhole x3
- Library of Babble
- Mother x2
- Quixo The Adventurer
- Timetraveler

Shadows:

- Ghostly Hand
- Miasma
- One Last Job
- Oubliette
- Pawn Sacrifice
- Relentless Whispers x3
- Too Much to Protect
- Bad Penny
- Old Bruno
- Duskrunner

Cards

Currently in Phase 1

If multiple effects are given with only one trigger like -> Play Archive 1 Card, Discard 1 Card -> that means that the trigger activates all those effects, but the effects are resolved in order. In this example, the player must first archive 1 card before they discard 1 card.

Dis

Arise

Æmber on Play: 0
ID: 94
Tags: None
Type: Action
Effects: Play Arise

Control the Weak

Æmber on Play: 1
ID: 55
Tags: None
Type: Action
Effects: Play Control the Weak

Creeping Oblivion

Æmber on Play: 1
ID: 56
Tags: None
Type: Action
Effects: Play Purge 2 Cards from any Discard Pile

Dominator Bauble

Æmber on Play: 0
ID: 73
Tags: Item
Type: Artifact
Effects: Action Dominator Bauble

Dust Imp

Æmber on Play: 0
ID: 83
Tags: Imp
Type: Creature
Power: 2
Armor: 0
Effects: Destroyed Gain 2 Æmber

Ember Imp

Æmber on Play: 0

ID: 85

Tags: Imp

Type: Creature

Effects: Passive Ember Imp

Gateway to Dis

Æmber on Play: 0

ID: 59

Tags: None

Type: Action

Effects: Play Gateway to Dis

Guardian Demon

Æmber on Play: 0

ID: 84

Tags: Demon

Type: Creature

Power: 4

Armor: 0

Effects: Play/After Fight/After Reap Guardian Demon

Lash of Broken Dreams

Æmber on Play: 0

ID: 75

Tags: Weapon

Type: Artifact

Effects: Action Lash of Broken Dreams

Library of the Damned

Æmber on Play: 0

ID: 76

Tags: Location

Type: Artifact

Effects: Action Archive a Card

Lifeward

Æmber on Play: 1

ID: 77

Tags: Power

Type: Artifact

Effects: Omni Lifeward

Pit Demon

Æmber on Play: 0

ID: 92

Tags: Demon

Type: Creature

Power: 5

Armor: 0

Effects: Action Steal 1 Æmber

Shooler

Æmber on Play: 0

ID: 96

Tags: Demon

Type: Creature

Power: 5

Armor: 0

Effects: Play Shooler

Snudge

Æmber on Play: 0

ID: 97

Tags: Demon

Type: Creature

Power: 4

Armor: 0

Effects: After Fight/After Reap Snudge

Succubus

Æmber on Play: 0

ID: 99

Tags: Demon

Type: Creature

Power: 3

Armor: 0

Effects: Passive CardDrawModifier Enemy - 1

The Terror

Æmber on Play: 0

ID: 101

Tags: Demon, Knight

Type: Creature

Power: 5

Armor: 0

Effects: Play The Terror

Three Fates

Æmber on Play: 0

ID: 88

Tags: None

Type: Action

Effects: Play Three Fates

Logos

Doc Bookton

Æmber on Play: 0

ID: 101

Tags: Human, Scientist

Type: Creature

Power: 5

Armor: 0

Effects: After Reap Draw 1 Card

Help From Future Self

Æmber on Play: 1

ID: 111

Tags: None

Type: Action

Effects: Play Help From Future Self

Labwork

Æmber on Play: 1

ID: 114

Tags: None

Type: Action

Effects: Play Archive 1 Card

Library Access

Æmber on Play: 0

ID: 317

Tags: None

Type: Action

Effects: Play Library Access

Library of Babble

Æmber on Play: 0
ID: 129
Tags: Location
Type: Artifact
Effects: Action Play 1 Card

Mother

Æmber on Play: 0
ID: 156
Tags: Robot, Scientist
Type: Creature
Power: 5
Armor: 0
Effects: Passive Mother

Phase Shift

Æmber on Play: 0
ID: 157
Tags: None
Type: Action
Effects: Play Phase Shift

Quixo the Adventurer

Æmber on Play: 0
ID: 144
Tags: Human, Scientist
Type: Creature
Power: 3
Armor: 0
Effects: Fight Draw 1 Card, Passive Skirmish

Scrambler Storm

Æmber on Play: 1
ID: 122
Tags: None
Type: Action
Effects: Play Scrambler Storm

Sloppy Labwork

Æmber on Play: 1
ID: 123
Tags: None
Type: Action
Effects: Play Archive 1 Card, Discard 1 Card

The Howling Pit

Æmber on Play: 1

ID: 135

Tags: Location

Type: Artifact

Effects: Passive CardDrawModifier Player, Enemy + 1

Timetraveler

Æmber on Play: 1

ID: 153

Tags: Human, Scientist

Type: Creature

Power: 2

Armor: 0

Effects: Play Draw 2 Cards, Action Shuffle This Into Deck

Titan Mechanic

Æmber on Play: 0

ID: 154

Tags: Cyborg, Scientist

Type: Creature

Power: 6

Armor: 0

Effects: Passive Titan Mechanic

Wild Wormhole

Æmber on Play: 1

ID: 125

Tags: None

Type: Action

Effects: Play Wild Wormhole

Shadows

Bad Penny

Æmber on Play: 0

ID: 296

Tags: Human, Thief

Type: Creature

Power: 1

Armor: 0

Effects: Destroyed Bad Penny

Bait and Switch

Æmber on Play: 0

ID: 267

Tags: None

Type: Action

Effects: Play Bait and Switch

Booby Trap

Æmber on Play: 1

ID: 268

Tags: None

Type: Action

Effects: Play Booby Trap

Duskrunner

Æmber on Play: 0

ID: 316

Tags: None

Type: Upgrade

Effects: Reap Steal 1 Æmber

Ghostly Hand

Æmber on Play: 2

ID: 270

Tags: None

Type: Action

Effects: Play Ghostly Hand

Lights Out

Æmber on Play: 1

ID: 274

Tags: None

Type: Action

Effects: Play 2 Enemy Creatures to Hand

Miasma

Æmber on Play: 1

ID: 275

Tags: None

Type: Action

Effects: Play Skip Enemy KeyForge 2

Nerve Blast

Æmber on Play: 0

ID: 276
Tags: None
Type: Action
Effects: Play Nerve Blast

Noddy the Thief

Æmber on Play: 0
ID: 306
Tags: Elf, Thief
Type: Creature
Power: 2
Armor: 0
Effects: Passive Elusive, Action Steal 1 Æmber

Old Bruno

Æmber on Play: 0
ID: 307
Tags: Elf, Thief
Type: Creature
Power: 3
Armor: 0
Effects: Play Capture 3 Æmber, Passive Elusive

One Last Job

Æmber on Play: 1
ID: 277
Tags: None
Type: Action
Effects: Play One Last Job

Oubliette

Æmber on Play: 0
ID: 278
Tags: None
Type: Action
Effects: Play Purge Creature with power 3 or less from Play Area

Pawn Sacrifice

Æmber on Play: 1
ID: 279
Tags: None
Type: Action
Effects: Play Pawn Sacrifice

Relentless Whispers

Æmber on Play: 1

ID: 281

Tags: None

Type: Action

Effects: Play Relentless Whispers

Silvertooth

Æmber on Play: 0

ID: 311

Tags: Elf, Thief

Type: Creature

Power: 2

Armor: 0

Effects: Play Readied Silvertooth

Subtle Maul

Æmber on Play: 0

ID: 294

Tags: Weapon

Type: Artifact

Effects: Action Enemy Discards 1 Card

Too Much To Protect

Æmber on Play: 1

ID: 283

Tags: None

Type: Action

Effects: Play Too Much To Protect

Urchin

Æmber on Play: 0

ID: 584

Tags: Thief, Elf

Type: Creature

Power: 1

Armor: 0

Effects: Play Steal 1 Æmber, Passive Elusive

Effects

Effects are either generic or named. Generic abilities may be used on multiple cards, and the numbers may vary. Generic abilities are named with the trigger(s) ahead of the ability, such as “Action Gain 1 Æmber”. Named are used only for the named card. Unless specified, the hand, discard pile, and so on mentioned in abilities are for the controlling player. Each effect is either duration, trigger, instead, or instantly resolved. If it is instantly resolved, it won’t be marked. Otherwise, it will be marked with parentheses after the effect name and will also be labeled with how long the effect should last in a number of turns. 1 end of turn means the end of the current turn, while 2 means until your next turn. This will be labeled like (Duration - 2) to indicate that the effect will be removed after two turns end. Variables in effects are labeled like [Players] which can be used to indicate which players are affected.

If a duration is marked infinite, that means the effect ends when the source of the effect is removed from the play area and continues until that happens. Duration effects run their steps immediately and then log the duration effect.

If unspecified, any “play card” type effect works as if all cards in the hand CanBePlayed. It is still superseded by specific limits like CanPlayCreatures = False, but not limited to the currently selected House.

If you cannot complete a step for an effect, then stop there and do not continue to execute the next steps.

Generic:

Archive X Card(s)

Steps:

1. Put X cards from the hand into the archive

Capture X Æmber

Steps:

1. Decrease the enemy Æmber pool count by X
2. Increase this Card’s Æmber Captured count by however much was successfully decreased in step 1.

Draw X Card(s)

Steps:

1. Draw X number of cards

[Player] Discards X Card(s)

Steps:

1. Put X card(s) from [Player]’s hand at random into the discard pile.

Elusive (Duration - infinite)

PlayerAffected: Player
Name of variable affected: OriginatingCard.Elusive
Value of variable affected: True

Gain X Æmber

Steps:

1. Increase the Æmber pool count by X

CardDrawModifier [Players] [+/-/=] X (Duration - infinite)

PlayerAffected: [Players]

Name of variable affected: CardDrawModifier

Value of variable affected: CurrentValue [+/-/=] X

Purge X Cards from Location

Steps:

1. Put X cards from location into the purged zone

Readied [Card]

Steps:

1. [Card]'s exhaustion is set to false

Return [Cards] to Hand

Steps:

1. Put [Cards] into owners hand

Shuffle [Card/ X in Location] Into Deck

Steps:

1. Select X cards in location, or locate the specified card
2. Shuffle them into the deck

Skip [Player] Keyforge [Duration] (Duration - [Duration])

PlayerAffected: [Player]

Name of variable affected: CanKeyForge

Value of variable affected: False

Skirmish

PlayerAffected: Player

Name of variable affected: OriginatingCard.Skirmish

Value of variable affected: True

Steal X Æmber

Steps:

1. Decrease the enemies Æmber pool count by X
2. If successful for Y, increase the Æmber pool count by Y

Named:

Arise:

Steps:

1. Choose a house you own
2. Put all Creature cards from the discard pile into your hand
3. Gain 1 chain

Bad Penny:

RemainingDuration: Instant (only takes effect now)

PlayerAffected: Player

Do: Put Bad Penny into hand

InsteadOf: Destroy Bad Penny

Bait and Switch:

Steps:

1. Decrease the enemies Æmber pool count by 1
2. If successful, increase the Æmber pool count by 1
3. Repeat from step 1 if your Æmber pool count is less than the enemy's Æmber pool count

Booby Trap

Steps:

1. Select a creature that is not on the Flank
2. Increase the damage counter on the selected Creature by 4
3. Increase the damage counter on the creature immediately before the selected Creature by 2
4. Increase the damage counter on the creature immediately after the selected Creature by 2

Control the Weak (Duration - 2)

Steps:

1. Choose a house the enemy owns

PlayerAffected: Enemy

Name of variable affected: HouseSelection

Value of variable affected: ChosenHouse

Dominator Bauble (Duration - 1)

Steps:

1. Select a creature you control

PlayerAffected: Player

Name of variable affected: SelectedCard.CanBeUsed

Value of variable affected: True

Ember Imp (Duration - infinite)

PlayerAffected: Enemy

Name of variable affected: CardPlayedLimit

Value of variable affected: 2

Gateway to Dis

Steps:

1. Destroy every creature in both play areas
2. Gain 3 chains

Ghostly Hand

Steps:

1. If opponents Æmber count is 1:
 - a. Decrease the enemies Æmber pool count by 1
 - b. If successful, increase the Æmber pool count by 1

Guardian Demon

Steps:

1. Remove up to 2 damage from any creatures damage counter, to a minimum of 0 total
2. For every damage removed, add an equal amount to another creature

Help From Future Self

Steps:

1. Search your deck for a Timetraveler card. If you don't find one, search your discard pile.
2. If one is found, show it to the opponent and then put it in your hand.
3. Put all cards from your discard pile into your deck
4. Shuffle your deck

Lash of Broken Dreams (Duration - 1)

PlayerAffected: Enemy

Name of variable affected: KeyForgeCost

Value of variable affected: CurrentValue plus 3

Library Access (Trigger - 1)

Steps:

1. Put Library Access in the Purged Zone

PlayerAffected: Player

Do: Draw a card

When: Play a card

Lifeward (Duration - 2)

Steps:

1. Destroy Lifeward

PlayerAffected: Enemy

Name of variable affected: CanPlayCreatures

Value of variable affected: False

Mother (Duration - Infinite)

PlayerAffected: Player

Name of variable affected: DrawUpToLimit

Value of variable affected: CurrentValue plus 1

Nerve Blast

Steps:

1. Decrease the enemies Æmber pool count by 1
2. If this succeeded:
 - a. Increase the Æmber pool count by 1
 - b. Increase the damage counter on a creature by 2

One Last Job

Steps:

1. Purge each creature of House Shadows you control
2. For X creatures purged
 - a. Decrease the enemies Æmber pool count by 1
 - b. If successful, increase the Æmber pool count by 1

Pawn Sacrifice:

Steps:

1. Destroy a creature you control
2. If a creature is destroyed:
 - a. Increase the damage counter on each of the two creatures by 3

Phase Shift (Duration - 1)

PlayerAffected: Player

Name of variable affected: NonLogosCardsPlayable

Value of variable affected: 1

Relentless Whispers

Steps:

1. Increase the damage counter on a creature by 2
2. Check if that creature is destroyed. If it is:
 - a. Decrease the enemies Æmber pool count by 1
 - b. If successful, increase the Æmber pool count by 1

Scrambler Storm (Duration - 1)

PlayerAffected: Enemy

Name of variable affected: CanPlayActions

Value of variable affected: False

Shooter

Steps:

1. If your opponent has at least 4 Æmber, then:
 - a. Decrease the enemies Æmber pool count by 1
 - b. If successful, increase the Æmber pool count by 1

Snudge

Steps:

1. Select an artifact or a creature on the Flank
2. Return the selected card to it's owner hand

The Terror

Steps:

1. If your opponent has no Æmber then:
 - a. Increase the Æmber pool count by 2

Three Fates

Steps:

1. Rank the power of the creatures in both play areas
2. Destroy the 3 with the most power. If there are ties, the player chooses which to destroy among the tied creatures.

Titan Mechanic (Duration - infinite)

PlayerAffected: Player, Enemy

Conditional: If Titan Mechanic is Flank

Name of variable affected: KeyForgeCost

Value of variable affected: CurrentValue minus 1

Too Much To Protect

Steps:

1. For $X = \max(\text{Enemy } \text{Æmber} - 6, 0)$
2. Decrease the enemies Æmber pool count by X
3. If successful for Y, increase the Æmber pool count by Y

Wild Wormhole:

Steps:

1. Take the top card from the deck
2. Play it and resolve it's effect

Variables

Player

CanKeyForge = True

Whether or not a key can be forged if there is enough Æmber to do so. If false, no key is forged regardless of how much Æmber there is.

CanPlayActions = True

When set to false, even if there are actions in their hand that CanBeUsed, they may not use them.

CanPlayCreatures = True

When set to false, even if there are creatures in their hand that CanBeUsed, they may not use them.

CardDrawModifier = 0

Added to the number of cards the player would draw in whatever draw action. If 0 or less, no cards are drawn.

CardPlayedLimit = null

By default, the player can play any card in their hand that CanBeUsed. This is indicated by having the default value here be null. When set to an integer though, this restricts the number of cards the player can play from their hand in a given turn. For example, if set to 2, then the player may only play a total of 2 cards from their hand in a turn. This does not affect their ability to use other cards and does not affect any effects that may put cards in other places.

CardsPlayed = []

A map of all cards in your deck plus any you control and the number of times each has been used. When one is used, increment the number of times that card has been used. Used to track how many cards have been played of each house, how many total, and used to ensure the rule of 6. At the end of the turn, this is reset.

Chains = 0

Max of 24, minimum of 0. Everytime someone with n chains would draw cards during step 5 of the turn order they draw $\text{math.ceil}(n/6)$ fewer cards and reduce their chains by 1. If they draw cards otherwise they are not affected by chains.

DrawUpToLimit = 6

At the end of each turn, a player draws the number of cards they would need to reach this number in their hand. For example, if they have 4 cards, and this is 6, then they would draw 2 cards. The amount they draw can be affected by the CardDrawModifier, but this determines the initial draw amount. If the user has cards equal to or higher than

this number they do not draw cards at the end of the round. This is also the starting hand size.

KeyForgeCost = 6

The number of Æmber needed at the beginning of a turn to forge a key. If there is at least as much Æmber as this number at the start of a players turn, subtract this number from the Æmber Pool count and increase the number of Keys for that player by 1. If this is their third Key, they win.

NonLogosCardsPlayable = True

When set to false, the user may not play Logos cards from their hand even if the card CanBeUsed. When set to true, the player can use Logos cards if they CanBeUsed.

HouseSelection = null

The house selected by the player each turn. Cleared at the end of every turn unless an effect is setting it. If null at the beginning of turn, the player chooses one of three houses that are in their deck (for Phase 1 and 2, this is Shadows, Dis, and Logos). Once chosen, cards of that house have CanBeUsed set to True.

Card

Originating Card - The card that started the effect

Selected Card - The card selected by the player

Elusive = False

If set to true, then the first time on any turn when this card is involved in a fight, skip the fight phase of the fight and proceed directly from the before fight phase to the after fight phase. Can be overwritten by the IgnoreElusive variable.

IgnoreElusive = False

If true, then regardless of whether or not isElusive is true, the fight proceeds as if elusive is false.

Skirmish = False

If true, then the creature with this set to true does not take damage from the other creature during the fight phase (their damage counter remains static). Other effects can still cause it to take damage during the fight phase, but the damage that is caused simply due to the fight don't occur. The creature still deals damage during the fight phase (the enemies damage counter increases as normal).

CanBeUsed = False

If this is true, then the card can be selected by the player to be played, to use an action effect, to reap or fight if it is a creature, and so on. The Omni keyword allows a card to be used for a particular effect even when this is false.

Exhausted = False

If true, then even if the card CanBeUsed it can no longer be used. When a card is used, it generally changes this to True.

Time

9/9/26 - 5 hours - Drafting the implementation requirements

9/11 - 4 hours - Continuing to pseudocode and gather requirements and info

9/12 - 1 hour - Revising pseudocode